

MATH 3032 — Mathematical Statistics

Chapter 4: Hypothesis Testing

Deep Understanding Guide + Gaussian Quiz Prep

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Quiz focus. Your quiz is on Gaussian hypothesis testing. Parts VI and VII are the most important. Work every quiz problem in Part X before reading the solution. The quiz will likely require: writing the test function δ , computing the Type I error probability using Φ , computing the Type II error probability, and identifying the rejection region.

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Part I: Why We Test Hypotheses

Chapter 3 answered: *what is the best estimate $\hat{\theta}$ of θ ?* Chapter 4 asks something different: *given data, can we decide whether θ belongs to one set or another?*

The fundamental question shift

Estimation (Ch. 3): Find $\hat{\theta}$ close to θ .

Testing (Ch. 4): Given data, decide: is $\theta \in \Theta_0$ or $\theta \in \Theta_1$?

These are related but distinct. Testing is about *decision-making under uncertainty*, not just point estimation.

1.1. The Setup (from Lecture Slides)

Given $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F_\theta$, we want to decide between:

- H_0 : the **null hypothesis**, $\theta \in \Theta_0$ (the “default” or “no effect” claim)
- H_1 : the **alternative hypothesis**, $\theta \in \Theta_1$ (the “signal” or “effect” claim)

Convention: Choose H_0 as the hypothesis whose false rejection is more harmful.

Decision tool: A test function $\delta : \mathcal{X}^n \rightarrow \{0, 1\}$ where:

$$\delta = 1 \Leftrightarrow \text{reject } H_0 \text{ (decide in favor of } H_1), \quad \delta = 0 \Leftrightarrow \text{fail to reject } H_0.$$

In practice, δ is built from a **test statistic** $T(X_1, \dots, X_n)$ and a **critical region** C :

$$\delta(X_1, \dots, X_n) = \mathbf{1}_{T(X_1, \dots, X_n) \in C}.$$

δ is a **Bernoulli random variable**. Since it takes values in $\{0, 1\}$ and its value depends on the random sample, $\delta \sim \text{Ber}(p)$ where $p = P_\theta(T \in C)$. This p depends on the true θ — which is exactly what we’re trying to learn about.

Part II: Error Probabilities, Significance, and Power

2.1. The Error Table

Two mistakes are possible:

	Decide H_0 ($\delta = 0$)	Decide H_1 ($\delta = 1$)
Truth: H_0	No error	Type I error
Truth: H_1	Type II error	No error

Type I error probability (false positive rate):

$$h : \Theta_0 \rightarrow [0, 1], \quad h(\theta) = P_\theta[\delta = 1], \quad \theta \in \Theta_0.$$

Type II error probability (false negative rate):

$$g : \Theta_1 \rightarrow [0, 1], \quad g(\theta) = P_\theta[\delta = 0], \quad \theta \in \Theta_1.$$

Power = $1 - g(\theta) = P_\theta[\delta = 1]$ for $\theta \in \Theta_1$. Higher power = better test.

Critical warning from lecture: $h(\theta) \neq 1 - g(\theta)$. These functions have *different domains*. h is defined over Θ_0 and g over Θ_1 . They are not complements of each other. This is a common source of confusion on exams.

2.2. The Fundamental Trade-off

From lecture slide 3: if we shrink the critical region C to $C^* \subseteq C$:

- Type I error decreases: $P_\theta(T \in C^*) \leq P_\theta(T \in C)$ for $\theta \in \Theta_0$.
- Type II error increases: $P_\theta(T \notin C^*) \geq P_\theta(T \notin C)$ for $\theta \in \Theta_1$.

You cannot minimize both simultaneously. This is why we need a framework.

Part III: The Neyman-Pearson Framework

Idea: Fix the Type I error budget first, then minimize Type II error within that budget.

1. Fix $\alpha \in (0, 1)$ — the **significance level** (commonly 0.05 or 0.01).
2. Restrict attention to $\mathcal{D}(\Theta_0, \alpha)$, the class of test functions that *respect the level*:

$$\mathcal{D}(\Theta_0, \alpha) := \left\{ \delta : \mathcal{X}^n \rightarrow \{0, 1\} : \max_{\theta \in \Theta_0} h(\theta) \leq \alpha \right\}.$$

3. Among all $\delta \in \mathcal{D}(\Theta_0, \alpha)$, select the one that *minimizes* $g(\theta)$ (equivalently, *maximizes power*).

Intuition: $\alpha = 0.05$ means “I am willing to incorrectly reject H_0 at most 5% of the time when H_0 is true.” Given this constraint, find the test that is most likely to correctly reject H_0 when H_1 is actually true.

3.1. The Three Testing Regimes

From lecture slide 8, there are three types of hypothesis pairs we study:

Type	H_0	H_1	Key feature
Simple	$\theta = \theta_0$	$\theta = \theta_1$	Both are single points
Unilateral	$\theta \leq \theta_0$	$\theta > \theta_0$	One-sided
	$\theta \geq \theta_0$	$\theta < \theta_0$	One-sided (flipped)
Bilateral	$\theta = \theta_0$	$\theta \neq \theta_0$	Two-sided

Deep intuition for which to use:

- **Simple vs Simple:** You know exactly what both worlds look like. Use the NP Lemma. Optimal test always exists.
- **Unilateral:** You only care about deviations in one direction. E.g., testing if a new drug is *better* (not just different) than a placebo.
- **Bilateral:** You care about deviations in either direction. E.g., testing if a coin is unfair (could be biased toward heads *or* tails).

Part IV: Simple Testing — The Neyman-Pearson Lemma

4.1. Setup

$H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ with $\theta_0 \neq \theta_1$. Both hypotheses are single points.

Define the **likelihood ratio**:

$$\Lambda(\vec{X}) = \frac{f_{\theta_1}(\vec{X})}{f_{\theta_0}(\vec{X})} = \frac{L(\theta_1)}{L(\theta_0)}.$$

If there exists $Q > 0$ such that $P_{\theta_0}(\Lambda > Q) = \alpha$, then the test

$$\delta(\vec{X}) = \mathbf{1}_{\Lambda(\vec{X}) > Q}$$

is **optimal at level α** : it has minimal Type II error probability among all tests at level α .

Intuition: The likelihood ratio Λ asks: “How much more likely is the data under H_1 than under H_0 ?” If Λ is large, the data strongly favors H_1 , so we reject H_0 . The threshold Q is chosen to keep the Type I error at exactly α .

4.2. Worked Example: Exponential Testing (Lecture slides 11–13)

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$, test $H_0 : \lambda = \lambda_0$ vs $H_1 : \lambda = \lambda_1$ with $\lambda_0 < \lambda_1$.

Step 1: Likelihood ratio.

$$L(\lambda) = \lambda^n e^{-\lambda \sum X_i}, \quad \Lambda = \frac{L(\lambda_1)}{L(\lambda_0)} = \left(\frac{\lambda_1}{\lambda_0}\right)^n e^{-(\lambda_1 - \lambda_0) \sum X_i}.$$

Step 2: Simplify. Λ is a *decreasing* function of $T = \sum X_i$ (since $\lambda_1 > \lambda_0$, the exponent is negative times T). So:

$$\Lambda > Q \iff T \leq q_\alpha \quad \text{for some threshold } q_\alpha.$$

Step 3: Find q_α . Under H_0 , $T = \sum X_i \sim \text{Gamma}(n, \lambda_0)$. Choose:

$$q_\alpha = \alpha\text{-quantile of } \text{Gamma}(n, \lambda_0), \quad \text{i.e. } P_{\lambda_0}(T \leq q_\alpha) = \alpha.$$

Step 4: Test function.

$$\delta = \mathbf{1}_{T \leq q_\alpha} = \mathbf{1}_{\sum X_i \leq q_\alpha}$$

Reject H_0 when the sample sum is unusually *small* (high λ_1 means shorter waiting times).

Numerical example from lecture: $\lambda_0 = 1$, $\lambda_1 = 2$, $n = 100$, $\alpha = 0.05$.
 $q_\alpha \approx 84.14$ (5th percentile of $\text{Gamma}(100, 1)$).
 Reject H_0 whenever $\sum X_i < 84.14$, i.e. when $\Lambda > 0.000000364$.

Part V: Unilateral (One-Sided) Testing

5.1. The Setup and General Result

$H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$ (or flipped). Here Θ_0 is a *set* (a ray), not a single point.

For pdfs in exponential family form $f_\theta(x) = e^{\eta(\theta)K(x) - d(\theta) + S(x)}$, use sufficient statistic $\tau = \sum K(X_i)$.

	$H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$	$H_0 : \theta \geq \theta_0$ vs $H_1 : \theta < \theta_0$
$\eta(\cdot)$ increasing	$\mathbf{1}\{\tau > q_{1-\alpha}\}$	$\mathbf{1}\{\tau \leq q_\alpha\}$
$\eta(\cdot)$ decreasing	$\mathbf{1}\{\tau \leq q_\alpha\}$	$\mathbf{1}\{\tau > q_{1-\alpha}\}$

5.2. How to Read the Table

Key insight: Whether the likelihood ratio is increasing or decreasing in the sufficient statistic τ determines the direction of the rejection region. Here is the logic:

- For $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$: Evidence against H_0 is large θ , which corresponds to large or small τ depending on whether $\eta(\theta)$ is increasing or decreasing in θ .
- **η increasing:** Large $\theta \rightarrow$ large $\eta(\theta) \rightarrow$ large τ favors $H_1 \rightarrow$ reject when $\tau > q_{1-\alpha}$.
- **η decreasing:** Large $\theta \rightarrow$ small $\eta(\theta) \rightarrow$ small τ favors $H_1 \rightarrow$ reject when $\tau \leq q_\alpha$.

5.3. Exponential Example (Lecture slides 20–23)

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$, test $H_0 : \lambda \leq \lambda_0$ vs $H_1 : \lambda > \lambda_0$.

The pdf is $f_\lambda(x) = \lambda e^{-\lambda x}$, which in exponential family form is $f_\lambda(x) = e^{\log \lambda + (-\lambda)x}$.

So $\eta(\lambda) = -\lambda$ (decreasing in λ), $K(x) = x$, sufficient statistic $\tau = \sum X_i$.

Since η is *decreasing* and we have $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$: reject when $\tau \leq q_\alpha$.

Under H_0 evaluated at the boundary $\lambda = \lambda_0$: $T \sim \text{Gamma}(n, \lambda_0)$.

$$q_\alpha : P_{\lambda_0}(T \leq q_\alpha) = \alpha \quad \Rightarrow \quad \delta = \mathbf{1}_{T \leq q_\alpha}.$$

Why this is optimal for ALL $\lambda_1 > \lambda_0$: The test $\delta = \mathbf{1}_{T \leq q_\alpha}$ can be rewritten as $\mathbf{1}_{\Lambda > Q(\lambda_1)}$ for any $\lambda_1 > \lambda_0$. By the NP Lemma applied pointwise, no other test at level α can beat it for any specific λ_1 . Hence it's **uniformly most powerful** (UMP).

Part VI: Bilateral (Two-Sided) Gaussian Testing

6.1. Setup — Known Variance (Lecture slides 27–30)

This is the **most important section for your quiz**.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ^2 **known**. Test $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$.

Why bilateral? Unlike unilateral tests, we care about deviations in both directions. If $\mu > \mu_0$, we want to reject; but if $\mu < \mu_0$, we also want to reject.

6.1.1. Step 1: Form the likelihood ratio

MLE is $\hat{\mu} = \bar{X}$. The likelihood ratio is:

$$\Lambda = \frac{L(\hat{\mu})}{L(\mu_0)} = \exp\left(\frac{n}{2\sigma^2}(\bar{X} - \mu_0)^2\right).$$

Using the algebraic identity (from lecture slide 28):

$$\sum (X_i - \mu_0)^2 = \sum (X_i - \bar{X})^2 + n(\bar{X} - \mu_0)^2.$$

6.1.2. Step 2: Identify the pivotal quantity

Λ is an increasing function of:

$$M := \left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}\right)^2.$$

So $\Lambda > Q \iff M > q_{1-\alpha}$ for some threshold $q_{1-\alpha}$.

6.1.3. Step 3: Identify the distribution under H_0

Under H_0 , $\mu = \mu_0$, so:

$$\bar{X} \sim \mathcal{N}\left(\mu_0, \frac{\sigma^2}{n}\right) \Rightarrow Z := \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1) \Rightarrow M = Z^2 \sim \chi^2(1).$$

6.1.4. Step 4: Find the critical value and test function

$$\alpha = P(M > q_{1-\alpha}) = P(\chi^2(1) > q_{1-\alpha}).$$

Since $\chi^2(1) > c \iff |Z| > \sqrt{c}$ and $P(|Z| > \sqrt{c}) = \alpha$:

$$2P(Z > z_{1-\alpha/2}) = \alpha \iff P(Z > z_{1-\alpha/2}) = \frac{\alpha}{2} \iff z_{1-\alpha/2} = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right).$$

$$\delta = \mathbf{1}_{\left|\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}\right| > z_{1-\alpha/2}}$$

where $z_{1-\alpha/2} = \Phi^{-1}(1 - \alpha/2)$ is the $(1 - \alpha/2)$ quantile of $\mathcal{N}(0, 1)$.

For $\alpha = 0.05$: $z_{0.975} = 1.96$. For $\alpha = 0.01$: $z_{0.995} = 2.576$.

6.2. Type I and Type II Error Probabilities for Gaussian Testing

6.2.1. Type I error probability

The test statistic under H_0 is $Z = (\bar{X} - \mu_0)/(\sigma/\sqrt{n}) \sim \mathcal{N}(0, 1)$.

$$h(\mu_0) = P_{\mu_0}(\delta = 1) = P(|Z| > z_{1-\alpha/2}) = \alpha. \quad \checkmark$$

6.2.2. Type II error probability

Under H_1 with true mean $\mu \neq \mu_0$: $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$, so $Z \sim \mathcal{N}(\mu\sqrt{n}/\sigma - \mu_0\sqrt{n}/\sigma, 1) = \mathcal{N}(\Delta, 1)$ where $\Delta = (\mu - \mu_0)\sqrt{n}/\sigma$.

$$\begin{aligned} g(\mu) &= P_\mu(\delta = 0) = P_\mu(|Z'| < z_{1-\alpha/2}) \quad \text{where } Z' = Z - \Delta \sim \mathcal{N}(0, 1) \\ &= P(-z_{1-\alpha/2} < Z' + \Delta < z_{1-\alpha/2}) \\ &= P(-z_{1-\alpha/2} - \Delta < Z' < z_{1-\alpha/2} - \Delta) \\ &= \Phi(z_{1-\alpha/2} - \Delta) - \Phi(-z_{1-\alpha/2} - \Delta). \end{aligned}$$

Type II error for bilateral Gaussian test:

$$g(\mu) = \Phi\left(z_{1-\alpha/2} - \frac{(\mu - \mu_0)\sqrt{n}}{\sigma}\right) - \Phi\left(-z_{1-\alpha/2} - \frac{(\mu - \mu_0)\sqrt{n}}{\sigma}\right)$$

Power = $1 - g(\mu)$. Power increases as $|\mu - \mu_0|$ increases (easier to detect larger effects) or as n increases (more data).

6.3. The Unilateral Gaussian Tests

Right-sided: $H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$

$$\delta = \mathbf{1}_{\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} > z_{1-\alpha}}, \quad h(\mu_0) = P(Z > z_{1-\alpha}) = \alpha.$$

Left-sided: $H_0 : \mu \geq \mu_0$ vs $H_1 : \mu < \mu_0$

$$\delta = \mathbf{1}_{\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} < -z_{1-\alpha}}, \quad h(\mu_0) = P(Z < -z_{1-\alpha}) = \alpha.$$

Key differences from bilateral:

- Critical value is $z_{1-\alpha}$ (NOT $z_{1-\alpha/2}$) — no factor of 2 because rejection is one-directional.
- For $\alpha = 0.05$: $z_{0.95} = 1.645$ (unilateral) vs $z_{0.975} = 1.96$ (bilateral).
- Unilateral tests are more *powerful* when the true alternative is in the right direction, because the critical value is lower.

Part VII: Unknown Variance — The Student t -Test

7.1. Why We Need a Different Test

In practice, σ^2 is rarely known. If we plug in $\hat{\sigma}^2$ for σ^2 , the statistic $(\bar{X} - \mu_0)/(\hat{\sigma}/\sqrt{n})$ is no longer $\mathcal{N}(0, 1)$ — it has heavier tails due to estimation uncertainty.

Sample variance (unbiased):

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

t-statistic (lecture slide 35):

$$T := \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, then under $H_0 : \mu = \mu_0$:

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1},$$

the Student's t -distribution with $n - 1$ degrees of freedom and pdf:

$$f_k(x) = \frac{\Gamma(\frac{k+1}{2})}{\Gamma(\frac{k}{2}) \sqrt{k\pi}} \left(1 + \frac{x^2}{k}\right)^{-\frac{k+1}{2}}, \quad k = n - 1.$$

For $k > 2$: $\mathbb{E}[T] = 0$ and $\text{Var}(T) = k/(k - 2) > 1$.

$H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$, unknown σ^2 :

$$\delta = \mathbf{1}_{|T| > t_{n-1; 1-\alpha/2}}$$

where $t_{n-1; 1-\alpha/2}$ is the $(1 - \alpha/2)$ quantile of the t_{n-1} distribution.

Why $|T| > c$? By symmetry of the t -distribution (symmetric around 0, just like $\mathcal{N}(0, 1)$), $P(|T| > c) = 2P(T > c) = \alpha$ gives $c = t_{n-1; 1-\alpha/2}$.

7.2. t -test vs Z -test: When to Use Which

	Z-test	t-test
Variance	Known: σ^2 given	Unknown: estimate with S^2
Test stat	$Z = (\bar{X} - \mu_0)/(\sigma/\sqrt{n})$	$T = (\bar{X} - \mu_0)/(S/\sqrt{n})$
Null dist	$\mathcal{N}(0, 1)$	t_{n-1}
Critical value	$z_{1-\alpha/2} = \Phi^{-1}(1 - \alpha/2)$	$t_{n-1; 1-\alpha/2}$
Large n behavior	Same	$t_{n-1} \rightarrow \mathcal{N}(0, 1)$, so $t \approx z$

Practical rule: The t -distribution has heavier tails than $\mathcal{N}(0, 1)$, so $t_{n-1; 1-\alpha/2} > z_{1-\alpha/2}$. This means the t -test requires more extreme data to reject H_0 , reflecting the additional uncertainty from estimating σ . As $n \rightarrow \infty$, $t_{n-1; 1-\alpha/2} \rightarrow z_{1-\alpha/2}$, so for large samples it doesn't matter much.

Part VIII: Nuisance Parameters

When the model has two parameters (θ, ξ) but we only care about testing θ , the parameter ξ is called a **nuisance parameter**.

Modified likelihood ratio (generalized LRT):

$$\Lambda = \frac{\max_{\theta \neq \theta_0, \xi} L(\theta, \xi)}{\max_{\xi} L(\theta_0, \xi)} = \frac{L(\hat{\theta}, \hat{\xi})_{\text{MLE}}}{\max_{\xi} L(\theta_0, \xi)}.$$

The bilateral t -test is exactly this setup: $\theta = \mu$ is of interest, $\xi = \sigma^2$ is the nuisance parameter. The denominator maximizes over σ^2 for fixed μ_0 ; the numerator uses the full MLE.

Part IX: Deep Intuition — How to Think About All Three Types

9.1. The Mental Model

Think of each test as asking a question about where your data falls relative to μ_0 :

All tests share the same test statistic:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \quad (\text{or } T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \text{ if } \sigma \text{ unknown})$$

The *only* difference is what counts as “extreme enough to reject”:

Test type	Reject H_0 when	Intuition
Bilateral ($\mu \neq \mu_0$)	$ Z > z_{1-\alpha/2}$	Too far from μ_0 in either direction
Right-sided ($\mu > \mu_0$)	$Z > z_{1-\alpha}$	Too far above μ_0
Left-sided ($\mu < \mu_0$)	$Z < -z_{1-\alpha}$	Too far below μ_0

9.2. Why $z_{1-\alpha}$ for Unilateral but $z_{1-\alpha/2}$ for Bilateral?

The α -budget. For $\alpha = 0.05$:

- **Bilateral:** split the budget equally between the two tails: $\alpha/2 = 0.025$ in each tail. So $z_{0.975} = 1.96$.
- **Unilateral:** put all the budget in one tail: $\alpha = 0.05$ in one tail. So $z_{0.95} = 1.645$.

Result: $1.645 < 1.96$, so right-sided tests reject more easily (lower bar) when $\mu > \mu_0$, but cannot detect $\mu < \mu_0$ at all.

9.3. The Φ function — Your Main Calculation Tool

$\Phi(x) = P(Z \leq x)$ for $Z \sim \mathcal{N}(0, 1)$. Key properties:

- $\Phi(-x) = 1 - \Phi(x)$ (symmetry)
- $P(Z > x) = 1 - \Phi(x)$
- $P(|Z| > x) = 2(1 - \Phi(x)) = 2\Phi(-x)$

- $P(a < Z < b) = \Phi(b) - \Phi(a)$

Critical values you must memorize:

α	$z_{1-\alpha}$ (unilateral)	$z_{1-\alpha/2}$ (bilateral)	$t_{n-1;1-\alpha/2}$ (large n)
0.10	1.282	1.645	≈ 1.645
0.05	1.645	1.960	≈ 1.960
0.01	2.326	2.576	≈ 2.576

Part X: Gaussian Hypothesis Testing Quiz — 15 Problems

These 15 problems cover everything your quiz will test. Work each problem completely before reading the solution. For Gaussian problems: (1) identify H_0 , H_1 , and test type; (2) write Z or T ; (3) state the rejection region; (4) compute Type I error in terms of Φ ; (5) compute Type II error if asked.

Notation: Φ = standard normal CDF. $z_p = \Phi^{-1}(p)$ is the p -quantile.

10.1. Warm-Up: Understanding the Setup

For each pair (H_0, H_1) , state whether it is Simple, Right-sided Unilateral, Left-sided Unilateral, or Bilateral:

- $H_0 : \mu = 5$ vs $H_1 : \mu = 7$
- $H_0 : \mu = 0$ vs $H_1 : \mu \neq 0$
- $H_0 : \mu \leq 3$ vs $H_1 : \mu > 3$
- $H_0 : \mu \geq 10$ vs $H_1 : \mu < 10$
- $H_0 : \sigma^2 = 4$ vs $H_1 : \sigma^2 \neq 4$ (variance test, not mean)

- Simple (both hypotheses are single points, $\theta_0 = 5$, $\theta_1 = 7$).
- Bilateral (H_1 is $\mu \neq 0$, two-sided).
- Right-sided unilateral ($H_1 : \mu > 3$).
- Left-sided unilateral ($H_1 : \mu < 10$).
- Bilateral for variance (same structure as bilateral mean test but for σ^2).

$X_1, \dots, X_{25} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, 16)$ (so $\sigma^2 = 16$, $\sigma = 4$). Test $H_0 : \mu = 10$ vs $H_1 : \mu \neq 10$ at level $\alpha = 0.05$.

- Write the test statistic Z explicitly.
- State the rejection region.
- Write the test function δ .
- What is the Type I error probability?

- (a) $Z = \frac{\bar{X} - 10}{4/\sqrt{25}} = \frac{\bar{X} - 10}{4/5} = \frac{5(\bar{X} - 10)}{4}$.
- (b) Bilateral test: reject when $|Z| > z_{0.975} = 1.96$.
- (c) $\delta = \mathbf{1}_{|Z| > 1.96} = \mathbf{1}_{\left|\frac{5(\bar{X} - 10)}{4}\right| > 1.96}$.
- (d) $h(\mu_0) = P_{\mu_0}(|Z| > 1.96) = 2(1 - \Phi(1.96)) = 2(0.025) = 0.05 = \alpha$. ✓

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$. Test $H_0 : \mu = 0$ vs $H_1 : \mu \neq 0$ using test $\delta = \mathbf{1}_{|\bar{X}| \geq Q}$.

- (a) Express $h(0) = P_{\mu=0}(\delta = 1)$ in terms of Φ , Q , n .
- (b) For $\alpha = 0.05$, find the smallest Q such that $h(0) \leq 0.05$.

This is exactly the computation from the lecture slides (Introduction to Hypothesis Testing, slides 8–9).

- (a) Under H_0 : $\bar{X} \sim \mathcal{N}(0, 1/n)$. Standardize: $U = \bar{X}/(1/\sqrt{n}) = \sqrt{n}\bar{X} \sim \mathcal{N}(0, 1)$.

$$h(0) = P_0(|\bar{X}| \geq Q) = P(|\bar{X}|/(1/\sqrt{n}) \geq Q\sqrt{n}) = P(|U| \geq Q\sqrt{n}) = 2\Phi(-Q\sqrt{n}).$$

- (b) Require $2\Phi(-Q\sqrt{n}) \leq 0.05$, i.e. $\Phi(-Q\sqrt{n}) \leq 0.025$, i.e. $-Q\sqrt{n} \leq \Phi^{-1}(0.025) = -1.96$. So $Q \geq 1.96/\sqrt{n}$. The smallest such Q is $Q = 1.96/\sqrt{n}$.

10.2. Core Calculations

$X_1, \dots, X_{100} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with $\sigma = 2$. Test $H_0 : \mu = 5$ vs $H_1 : \mu \neq 5$ at $\alpha = 0.05$.

- (a) Write the test function δ .
- (b) Compute $g(6) = P_{\mu=6}(\delta = 0)$ (Type II error when true mean is 6).
- (c) Compute the power at $\mu = 6$.

- (a) $Z = (\bar{X} - 5)/(2/\sqrt{100}) = (\bar{X} - 5)/0.2 = 5(\bar{X} - 5)$. $\delta = \mathbf{1}_{|Z| > 1.96}$.

- (b) Under $\mu = 6$: $\bar{X} \sim \mathcal{N}(6, 4/100)$, so $\bar{X} \sim \mathcal{N}(6, 0.04)$. Write $Z = 5(\bar{X} - 5)$; under $\mu = 6$: $Z \sim \mathcal{N}(5, 1)$ since $5(\mu - 5) = 5(1) = 5 = \Delta$.

$g(6) = P_6(|Z| \leq 1.96) = P_6(-1.96 \leq Z \leq 1.96)$. Let $Z' = Z - 5 \sim \mathcal{N}(0, 1)$:

$$g(6) = P(-1.96 - 5 \leq Z' \leq 1.96 - 5) = P(-6.96 \leq Z' \leq -3.04) = \Phi(-3.04) - \Phi(-6.96).$$

$\Phi(-3.04) \approx 0.0012$, $\Phi(-6.96) \approx 0$. So $g(6) \approx 0.0012$.

- (c) Power $= 1 - g(6) \approx 0.9988$. Very high — $\mu = 6$ is far from $\mu_0 = 5$ relative to $\sigma/\sqrt{n} = 0.2$.

$X_1, \dots, X_{36} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, 9)$. Test $H_0 : \mu \leq 0$ vs $H_1 : \mu > 0$ at $\alpha = 0.05$.

- (a) Write the test function δ .

- (b) Verify $\sup_{\mu \leq 0} h(\mu) \leq \alpha$.
 (c) Compute the power at $\mu = 1$.

(a) $\sigma = 3$, $\sigma/\sqrt{n} = 3/6 = 0.5$. $Z = (\bar{X} - 0)/0.5 = 2\bar{X}$. Right-sided: reject when $Z > z_{0.95} = 1.645$. $\delta = \mathbf{1}_{2\bar{X} > 1.645} = \mathbf{1}_{\bar{X} > 0.8225}$.
 (b) For any $\mu \leq 0$: $Z \sim \mathcal{N}(2\mu, 1)$ (mean ≤ 0). $h(\mu) = P_\mu(Z > 1.645) = P(\mathcal{N}(2\mu, 1) > 1.645) = 1 - \Phi(1.645 - 2\mu)$. Since $\mu \leq 0$: $1.645 - 2\mu \geq 1.645$, so $\Phi(1.645 - 2\mu) \geq \Phi(1.645) = 0.95$, giving $h(\mu) \leq 0.05$. ✓ Supremum achieved at $\mu = 0$: $h(0) = 1 - \Phi(1.645) = 0.05 = \alpha$.
 (c) At $\mu = 1$: $Z \sim \mathcal{N}(2, 1)$. Power $= P_1(Z > 1.645) = P(\mathcal{N}(2, 1) > 1.645) = 1 - \Phi(1.645 - 2) = 1 - \Phi(-0.355) = \Phi(0.355) \approx 0.639$.

$X_1, \dots, X_{16} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, 4)$. Test $H_0 : \mu \geq 3$ vs $H_1 : \mu < 3$ at $\alpha = 0.01$.

- (a) Write δ .
 (b) What is $h(3)$?
 (c) If observed $\bar{x} = 2.1$, do you reject H_0 ?

(a) $\sigma/\sqrt{n} = 2/4 = 0.5$. Left-sided: reject when $Z < -z_{0.99} = -2.326$. $Z = (\bar{X} - 3)/0.5 = 2(\bar{X} - 3)$. $\delta = \mathbf{1}_{2(\bar{X}-3) < -2.326} = \mathbf{1}_{\bar{X} < 3-1.163} = \mathbf{1}_{\bar{X} < 1.837}$.
 (b) $h(3) = P_3(Z < -2.326) = \Phi(-2.326) = 0.01 = \alpha$. ✓
 (c) $\bar{x} = 2.1$. Is $2.1 < 1.837$? No. Do not reject H_0 .

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$. Test $H_0 : \mu = 0$ vs $H_1 : \mu \neq 0$ at $\alpha = 0.05$. What is the minimum n such that power at $\mu = 0.5$ is at least 0.80?

Test statistic: $Z = \bar{X}\sqrt{n} \sim \mathcal{N}(0, 1)$ under H_0 ; $Z \sim \mathcal{N}(0.5\sqrt{n}, 1)$ under $\mu = 0.5$.
 Power at $\mu = 0.5$:

$$1 - g(0.5) = P_{\mu=0.5}(|Z| > 1.96) \approx P_{\mu=0.5}(Z > 1.96) = 1 - \Phi(1.96 - 0.5\sqrt{n}).$$

(The other tail $P(Z < -1.96)$ under $\mu = 0.5$ is negligible.)

Require $1 - \Phi(1.96 - 0.5\sqrt{n}) \geq 0.80$, i.e. $\Phi(1.96 - 0.5\sqrt{n}) \leq 0.20$.

$\Phi^{-1}(0.20) = -0.842$. So $1.96 - 0.5\sqrt{n} \leq -0.842$, giving $0.5\sqrt{n} \geq 2.802$, so $\sqrt{n} \geq 5.604$, thus $n \geq 32$ (round up).

Answer: $n = 32$.

10.3. The t -Test

$X_1, \dots, X_{10} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ^2 unknown. Observed $\bar{x} = 5.3$, $s^2 = 4$ (so $s = 2$). Test $H_0 : \mu = 4$ vs $H_1 : \mu \neq 4$ at $\alpha = 0.05$.

- Compute the t -statistic.
- State the rejection region using $t_{9;0.975} = 2.262$.
- Do you reject H_0 ?

- $T = (\bar{x} - \mu_0)/(s/\sqrt{n}) = (5.3 - 4)/(2/\sqrt{10}) = 1.3/(2/3.162) = 1.3/0.632 \approx 2.057$.
- Reject when $|T| > t_{9;0.975} = 2.262$.
- $|T| = 2.057 < 2.262$. Do NOT reject H_0 . (Just barely within the non-rejection region.)

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, σ^2 unknown. Test $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$ using the t -statistic $T = (\bar{X} - \mu_0)/(S/\sqrt{n})$.

- What is the distribution of T under H_0 ?
- Write the Type I error probability $h(\mu_0)$ in terms of the t -distribution CDF.
- Why is $t_{n-1;1-\alpha/2} > z_{1-\alpha/2}$?

- $T \sim t_{n-1}$ under H_0 .
- $h(\mu_0) = P_{H_0}(|T| > t_{n-1;1-\alpha/2}) = 2P(t_{n-1} > t_{n-1;1-\alpha/2}) = \alpha$. (By symmetry of t_{n-1} .)
- The t -distribution has heavier tails than $\mathcal{N}(0, 1)$ ($\text{Var}(T_k) = k/(k-2) > 1$). So the upper tail is thicker, meaning you need to go further out to capture the same probability in the tail. Hence the critical value is larger. As $n \rightarrow \infty$, $t_{n-1} \rightarrow \mathcal{N}(0, 1)$ and the values converge.

10.4. Challenge Problems

$X_1, \dots, X_m \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_X, \sigma^2)$ and $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_Y, \sigma^2)$ independent, σ^2 known. Test $H_0 : \mu_X = \mu_Y$ vs $H_1 : \mu_X \neq \mu_Y$.

- What is the distribution of $\bar{X} - \bar{Y}$ under H_0 ?
- Write the test statistic and rejection region.

- Under $H_0 : \mu_X = \mu_Y$: $\bar{X} - \bar{Y} \sim \mathcal{N}(0, \sigma^2/m + \sigma^2/n) = \mathcal{N}(0, \sigma^2(1/m + 1/n))$.
- $Z = \frac{\bar{X} - \bar{Y}}{\sigma\sqrt{1/m + 1/n}} \sim \mathcal{N}(0, 1)$ under H_0 . Reject when $|Z| > z_{1-\alpha/2}$.

$X_1, \dots, X_{25} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$. Bilateral test at $\alpha = 0.05$: reject when $|\bar{X}| > 0.392$.

- (a) Verify $0.392 = z_{0.975}/\sqrt{25}$.
- (b) Compute power at $\mu = 0.5$, $\mu = 1$, $\mu = 2$.
- (c) What happens to power as $|\mu| \rightarrow \infty$?

(a) $z_{0.975}/\sqrt{25} = 1.96/5 = 0.392$. ✓

(b) Power $= P_\mu(|\bar{X}| > 0.392) = P_\mu(|\bar{X} - \mu + \mu| > 0.392)$.

At $\mu = 0.5$: $Z = 5\bar{X} \sim \mathcal{N}(2.5, 1)$. Power $= P(|Z| > 1.96) = 1 - [\Phi(1.96 - 2.5) - \Phi(-1.96 - 2.5)] = 1 - \Phi(-0.54) + \Phi(-4.46) \approx 1 - 0.295 + 0 \approx 0.705$.

At $\mu = 1$: $Z \sim \mathcal{N}(5, 1)$. Power $\approx P(Z > 1.96) \approx 1 - \Phi(-3.04) \approx 0.999$.

At $\mu = 2$: $Z \sim \mathcal{N}(10, 1)$. Power ≈ 1 .

(c) Power $\rightarrow 1$. A consistent test detects any fixed alternative with probability approaching 1 as $|\mu - \mu_0| \rightarrow \infty$.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ^2 known. From Chapter 3, the asymptotic normality result gives:

$$\frac{\sqrt{n}(\hat{\mu} - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

- (a) Explain how this directly yields the Z -test for $H_0 : \mu = \mu_0$.
- (b) Why is the Z -test exact (not just approximate) for Gaussian data?
- (c) For non-Gaussian data with known variance, when can we still use the Z -test?

(a) Under $H_0 : \mu = \mu_0$, the asymptotic normality gives $\sqrt{n}(\bar{X} - \mu_0)/\sigma \xrightarrow{d} \mathcal{N}(0, 1)$. Use this as the test statistic; reject when $|Z| > z_{1-\alpha/2}$. This is exactly the Z -test.

(b) For Gaussian data, $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$ exactly (not just approximately). So $Z \sim \mathcal{N}(0, 1)$ exactly under H_0 , not just asymptotically. The Type I error is exactly α for any n .

(c) By the CLT, for large n , $\sqrt{n}(\bar{X} - \mu)/\sigma \approx \mathcal{N}(0, 1)$ regardless of the underlying distribution (as long as $\sigma^2 < \infty$). So the Z -test is approximately valid for any distribution with finite variance when n is large.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with known σ^2 . Write the likelihood ratio $\Lambda = L(\hat{\mu})/L(\mu_0)$ and show it simplifies to an increasing function of $M = (\bar{X} - \mu_0)^2/(\sigma^2/n)$. This is the derivation from lecture slides 27–28.

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i - \mu)^2}{2\sigma^2}\right) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum (X_i - \mu)^2}{2\sigma^2}\right).$$

$$\Lambda = \frac{L(\hat{\mu})}{L(\mu_0)} = \exp\left(-\frac{\sum (X_i - \hat{\mu})^2 - \sum (X_i - \mu_0)^2}{2\sigma^2}\right).$$

Using the identity $\sum (X_i - \mu_0)^2 = \sum (X_i - \bar{X})^2 + n(\bar{X} - \mu_0)^2$:

$$\sum (X_i - \hat{\mu})^2 - \sum (X_i - \mu_0)^2 = -n(\bar{X} - \mu_0)^2.$$

So:

$$\Lambda = \exp\left(\frac{n(\bar{X} - \mu_0)^2}{2\sigma^2}\right) = \exp\left(\frac{M}{2}\right),$$

where $M = n(\bar{X} - \mu_0)^2/\sigma^2$. Since $\exp(M/2)$ is increasing in M : $\Lambda > Q \iff M > c$ for some c . Under H_0 : $M = Z^2 \sim \chi^2(1)$. ✓

Blood pressure measurements: $X_1, \dots, X_{20} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, 25)$ (so $\sigma = 5$). A doctor claims the population mean is $\mu_0 = 120$ mmHg. You observe $\bar{x} = 124.3$.

- Test $H_0 : \mu = 120$ vs $H_1 : \mu \neq 120$ at $\alpha = 0.05$. Compute the test statistic, rejection region, and conclusion.
- Now test $H_0 : \mu \leq 120$ vs $H_1 : \mu > 120$ at $\alpha = 0.05$. Does your conclusion change?
- Compute the p -value for the bilateral test.
- If σ^2 were unknown and you observed $s = 5.2$, redo part (a) using the t -test. Use $t_{19;0.975} = 2.093$.

(a) Bilateral test: $Z = \frac{124.3 - 120}{5/\sqrt{20}} = \frac{4.3}{5/4.472} = \frac{4.3}{1.118} \approx 3.85$.

Rejection region: $|Z| > z_{0.975} = 1.96$. Since $3.85 > 1.96$, **reject** H_0 . Significant evidence that $\mu \neq 120$.

(b) Right-sided test: Rejection region: $Z > z_{0.95} = 1.645$. Since $3.85 > 1.645$, **reject** H_0 . Same conclusion, but this test is more powerful for detecting $\mu > 120$ specifically.

(c) p -value: $p\text{-value} = P(|Z| > 3.85) = 2(1 - \Phi(3.85)) \approx 2 \times 0.00006 = 0.00012$.

Extremely small — very strong evidence against H_0 .

(d) t -test: $T = (124.3 - 120)/(5.2/\sqrt{20}) = 4.3/(5.2/4.472) = 4.3/1.163 \approx 3.70$.

Rejection region: $|T| > t_{19;0.975} = 2.093$. Since $3.70 > 2.093$, **reject** H_0 . Same conclusion.

Note: the t -statistic (3.70) is slightly smaller than the Z -statistic (3.85) because $s = 5.2 > \sigma = 5$, making the denominator slightly larger.

Quick Reference Card

Bilateral Z -test (σ^2 known): $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1), \quad \delta = \mathbf{1}_{|Z| > z_{1-\alpha/2}}, \quad h(\mu_0) = \alpha.$$

Right-sided Z -test: $H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$

$$\delta = \mathbf{1}_{Z > z_{1-\alpha}}, \quad h(\mu_0) = \alpha.$$

Left-sided Z -test: $H_0 : \mu \geq \mu_0$ vs $H_1 : \mu < \mu_0$

$$\delta = \mathbf{1}_{Z < -z_{1-\alpha}}, \quad h(\mu_0) = \alpha.$$

Bilateral t -test (σ^2 unknown): $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}, \quad \delta = \mathbf{1}_{|T| > t_{n-1; 1-\alpha/2}}.$$

Type II error (bilateral, Z -test):

$$g(\mu) = \Phi(z_{1-\alpha/2} - \Delta) - \Phi(-z_{1-\alpha/2} - \Delta), \quad \Delta = \frac{(\mu - \mu_0)\sqrt{n}}{\sigma}.$$

p -value (bilateral): $p = 2(1 - \Phi(|Z_{\text{obs}}|)) = 2\Phi(-|Z_{\text{obs}}|)$.

p -value (right-sided): $p = 1 - \Phi(Z_{\text{obs}}) = \Phi(-Z_{\text{obs}})$.

Critical values: $z_{0.95} = 1.645$, $z_{0.975} = 1.96$, $z_{0.995} = 2.576$.

Exam checklist for hypothesis testing problems

- ☐ Identify H_0 and H_1 . Which type: simple, unilateral, bilateral?
- ☐ Is σ^2 known (Z -test) or unknown (t -test)?
- ☐ Write the test statistic (Z or T). Identify its distribution under H_0 .
- ☐ Determine the rejection region direction (both tails / right / left).
- ☐ Use $z_{1-\alpha/2}$ for bilateral, $z_{1-\alpha}$ for unilateral. Never mix these up.
- ☐ Verify $h(\mu_0) = \alpha$ by symmetry argument.
- ☐ For Type II error: shift the distribution by $\Delta = (\mu - \mu_0)\sqrt{n}/\sigma$ and recompute.
- ☐ State conclusion: reject H_0 / fail to reject H_0 . Never say “accept H_0 ”.